



Associations of individual nutrient and non-nutrient factors on endocrine and metabolic parameters in neonatal calves fed colostrum from their own dams

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ABSTRACT

Limited knowledge exists on the role of variation in individual colostrum components on metabolic and endocrine adaptation in calves. The objective of this study was to investigate the effects of individual colostral macronutrients and non-nutrient factors on metabolic and endocrine blood parameters of neonatal calves. Forty-six Holstein calves were fed with colostrum from their own dams at 4 and 12 h postpartum (p.p.). Blood samples were taken before feeding at 4, 12, and 24 h of age. Individual colostrum samples were analyzed for fat, protein, and lactose. Furthermore, concentrations of IgG, insulin, insulin-like growth factor (IGF)-1, and adiponectin were analyzed in colostrum. In blood samples, concentrations of IgG, total protein (TP), glucose, nonesterified fatty acids (NEFA), phospholipids (PL), triglycerides (TG), total cholesterol (TC), insulin, glucagon, IGF-1, and cortisol, as well as the activities of aspartate aminotransferase, and gamma-glutamyl transferase were measured. A mixed-effects model with time as fixed effect was used to evaluate temporal changes in blood parameters and to verify associations between colostral and plasma variables. In addition, Pearson correlation coefficients between colostrum and blood variables were calculated. We observed notable increases in plasma glucose, TP, and IgG concentrations. Plasma PL, TG, and TC concentrations were elevated only after 24 h p.p. Conversely, plasma NEFA concentration declined following initial colostrum intake. Significant correlations were observed between colostral fat content and plasma PL ($r = 0.65$ at 12 h p.p., $r = 0.76$ at 24 h p.p.), TG ($r = 0.51$ at 12 h p.p., $r = 0.54$ at 24 h p.p.), and TC ($r = 0.42$ at 12 h p.p., $r = 0.67$ at 24 h p.p.), which emphasizes the importance of colostral fat as an energy source. With regard to the immunization process, we identified associations between

colostral protein with plasma TP ($r = 0.45$ at 12 h p.p., $r = 0.49$ at 24 h p.p.) and IgG ($r = 0.56$ at 24 h p.p.); between colostral lactose content and plasma IgG ($r = -0.26$ at 12 h p.p.), TP ($r = -0.45$ at 12 h p.p., $r = -0.50$ at 24 h p.p.), and IgG ($r = -0.42$ at 24 h p.p.); between colostral IgG and serum IgG ($r = 0.28$ at 12 h p.p., $r = 0.62$ at 24 h p.p.) and TP ($r = 0.54$ at 24 h p.p.); between colostral insulin and serum IgG ($r = 0.52$ at 12 h p.p., $r = 0.40$ at 24 h p.p.) and TP ($r = 0.40$ at 12 h p.p., $r = 0.44$ at 24 h p.p.); and between colostral adiponectin and serum IgG ($r = 0.37$ at 12 h p.p.) and plasma TP ($r = 0.51$ at 12 h p.p.). In conclusion, distinct associations between colostrum components and the metabolic and immunological status of neonatal calves were observed when own-dam colostrum was fed. The correlations suggest that lactose, insulin, and adiponectin may influence IgG uptake, potentially through effects on intestinal development and closure, although these interpretations should be viewed as preliminary and require further confirmation.

Key words: calf, colostrum, energy metabolism, endocrine adaptation, immunization

INTRODUCTION

It is well known that the quality and quantity of colostrum available to newborn animals have significant influence on their subsequent health and well-being (Waltner-Toews et al., 1986; Wells et al., 1996; McGuirk and Collins, 2004). Elevated disease rates, developmental deficits, and mortality rates are commonly associated with poor colostrum quality or inadequate colostrum supply in the first hours of life (Wells et al., 1996; McGuirk and Collins, 2004; Raboisson et al., 2016). Undoubtedly, colostrum and the timely supply of IgG, respectively, are necessary to avoid failure in transfer of passive immunity (Fischer et al., 2018; Godden et al., 2019). Moreover, the initial 24 h likely represents a pivotal point in digestive physiology of the neonatal calf, marking the transition from uteroplacental nutrition via the dam to the nutrient digestion within a still-maturing gastrointestinal tract

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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

(GIT; Girard et al., 1992; Meale et al., 2017). In this context, colostrum and its bioactive components are substantially involved in postnatal GIT development and maturation (Baumrucker et al., 1994; Blättler et al., 2001; Hare et al., 2023). Colostrum contains both nutritive and non-nutritive factors (Silva et al., 2024). Nutritive factors include macronutrients such as protein, fat, and lactose, which provide energy and essential building blocks for the rapidly developing calf (Fischer et al., 2019; Silva et al., 2024). Non-nutritive, bioactive factors include hormones and growth factors such as insulin, insulin-like growth factor-1 (IGF-1), and adiponectin, which influence intestinal development, nutrient absorption, and early metabolic regulation (Baumrucker et al., 1994; Kesser et al., 2015; Hare et al., 2023). Nutrients must be absorbed within the first 24 h to compensate for the calf's postnatal hypoglycemic state (Girard et al., 1992; Hammon et al., 2013). Conversely, respective metabolic pathways such as endogenous glucose production (eGP) with gluconeogenesis (GNG) and glucose regulation need to be established within the same period (Scheuer et al., 2006). This metabolic transition is accompanied by rapid changes in blood metabolites within the first 24 h after birth. Blood glucose concentrations increase in response to colostrum ingestion, insulin is secreted in reaction to nutrient intake, and lipid metabolites such as triglycerides and phospholipids gradually appear in circulation (Hammon et al., 2000; Babitha et al., 2011; Fischer-Tlustos et al., 2021). Passive transfer of immunoglobulins, primarily IgG, also occurs during this period (Godden et al., 2009). These processes are highly dynamic and influenced by the composition and timing of colostrum intake. It can be reasonably deduced that ingested colostrum compounds play a pivotal role in ensuring energy supply and maturation of the offspring. Hence, a rapid and nutrient-dense colostrum supply enables sustaining of the calf's depleted glycogen reserves and facilitates the initiation of eGP (Okamoto et al., 1986; Hammon and Blum, 1998). In addition to macronutrients, numerous non-nutritive substances in colostrum such as insulin, adiponectin, and IGF-1 play an important role in GIT differentiation and affect energy metabolism (Baumrucker et al., 1994; Kesser et al., 2015; Hare et al., 2023). The variation in colostrum composition has been frequently addressed (Dunn et al., 2017; Gross et al., 2017; Kessler et al., 2020). However, limited knowledge exists regarding the role of variation in individual colostrum components on metabolic and endocrine adaptation in calves. Therefore, the objective of this study was to evaluate how the nutrient and non-nutrient composition of the individual colostrum from each calf's own dam, ingested during the first 24 h of life, influences their physiological and metabolic responses.

MATERIALS AND METHODS

The Commission for Animal Experimentation of the Canton of Fribourg (Switzerland) authorized experimental procedures in accordance with Swiss animal protection legislation (authorization numbers 2011-40-FR and 2013-18-FR).

Animals and Feeding

Data for this study originate from 46 neonatal Holstein calves (21 female, 25 male) from the Agroscope Research Station Posieux (Switzerland). Holstein dairy cows ($n = 46$; parity 3.5 ± 1.5 , range 2–7, $\pm$ SD) were enrolled at parturition, with all calvings occurring between late September and early March. Criteria for inclusion were absence of dystocia, no need for assisted calving, and singleton birth. Cows were dried off 8 wk before expected parturition, with BCS of 3.8 ± 0.9 ($\pm$ SD). Gestation length and dry period length were 283.5 ± 6.0 d and 69.7 ± 19.3 d ($\pm$ SD), respectively. Time of birth was continuously monitored around the clock by barn staff and a video camera positioned above the calving pens to ensure immediate detection. Calves were separated from their dams immediately after birth to prevent colostrum intake. Newborn calves were moved to individual boxes with straw bedding, where they remained for the initial 24 h postpartum (p.p.). Each calf was weighed after separation and monitored continuously during the first 24 h. Calves were fed colostrum (2 L) of their dam at 4 and 12 h p.p. The total colostrum intake per calf was 4 L, administered in two 2-L portions at 4 and 12 h p.p. Colostrum was completely consumed in almost all cases, with only 2 calves leaving small amounts (0.2 and 0.3 L) at 12 h. At 24 h p.p., bulk tank milk was fed to all calves.

Sampling and Analysis of Blood and Colostrum

Colostrum samples (~50 mL) were taken after milking cows within 4 h p.p. and frozen at -80°C until analysis. Blood samples were obtained from the calf's jugular vein (2×9 mL, Vacurette, Greiner Bio-One GmbH: K₃EDTA for plasma [cat. no. 455036] and CAT serum clot activator [cat. no. 455092] for serum separation) before the initial colostrum intake at 4 h p.p. (i.e., metabolic status before colostrum intake, onset of metabolic adaptations to extrauterine life; $n = 46$). Further samples were taken before second colostrum feeding at 12 h ($n = 46$) and before milk feeding at 24 h of age ($n = 21$). At 24 h p.p., only 21 calves could be sampled due to logistical constraints and availability of animals, leading to a reduced sample size at this point. This attrition is acknowledged and considered in the interpretation of results. Immediately following collection, blood samples for plasma

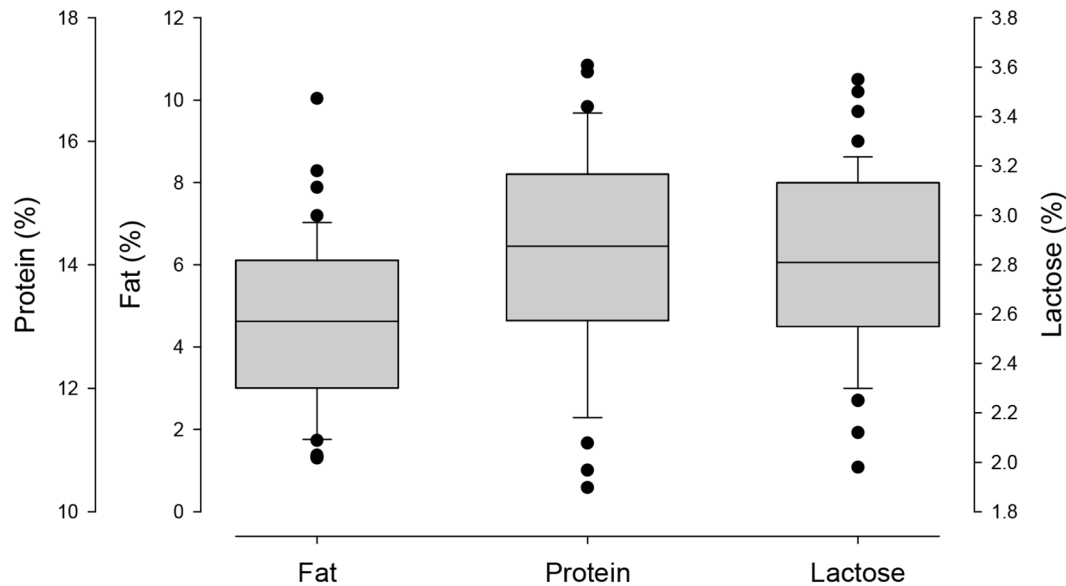


Figure 1. Concentrations of fat, protein, and lactose in colostrum fed to calves ($n = 46$). The lower and upper edges of the boxes represent the 25th and 75th percentiles, respectively; the midline indicates the median. Whiskers extend to the 10th and 90th percentiles, and dots outside these whiskers denote outliers.

harvest were cooled on ice and subsequently centrifuged ($1,500 \times g$ at 4°C for 20 min). Serum tubes were left at room temperature for 2 h to allow for further clotting and stored thereafter at 6°C until centrifugation. The collected plasma and serum were stored in 1.5-mL snap-vial tubes at -80°C until further analysis.

Adiponectin concentrations in colostrum could be measured only in a subset of cows whose calves were not sampled at 24 h p.p. Consequently, correlation analyses involving colostral adiponectin were restricted to the 4-h and 12-h blood samples, and no data were available for the 24-h time point. This limitation was considered when interpreting the potential influence of adiponectin on neonatal metabolic and immunological adaptation.

Contents of fat, protein, and lactose in colostrum samples were measured with a milk infrared analyzer in 1:2 diluted samples (MilkoScan 7 RM, Foss Analytical A/S, Hillerød, Denmark; analysis performed by Milchprüfing Bayern e.V., Wolnzach, Germany). Total IgG concentration in colostrum was determined with a modified ELISA kit (Bovine IgG ELISA Quantitation Set, cat. no. E10-118; Bethyl Laboratories Inc., Montgomery, TX) following the manufacturer's protocol and with slight modifications, as described by Lehmann et al., (2013), to final dilutions of 1:800,000 and 1:1,600,000. The interassay CV was 5.8%, and the intra-assay CV was 9.7%. Concentrations of insulin and IGF-1 in colostrum were determined by RIA, as described by Vicari et al. (2008). Adiponectin in colostrum was determined by ELISA at the Institute of Animal Science, Physiology and Hygiene Unit (Prof. Dr. H. Sauerwein, University of Bonn, Bonn,

Germany) as described in detail by Kesser et al. (2015). However, adiponectin concentrations could only be measured in a subset of colostrum from cows whose calves were not sampled at 24 h p.p. Consequently, correlation data with colostral adiponectin are restricted to the 12-h blood sample data.

Plasma concentrations of total protein (TP), glucose, nonesterified fatty acids (NEFA), phospholipids (PL), triglycerides (TG), and total cholesterol (TC), and activities of aspartate aminotransferase (ASAT) and gamma-glutamyl transferase (GGT) were determined by enzymatic assays using an autoanalyzer (Cobas Mira 2, Hoffman-La Roche Ltd., Basel, Switzerland) as specified by Freihofer et al. (2025). In serum, concentrations of insulin and IGF-1 were determined as described by Vicari et al. (2008). Serum concentration of glucagon was analyzed using a commercially available RIA kit (GL-32K; EMD Millipore), and cortisol was determined using an RIA with modifications as described by Schwinn et al. (2016). Serum concentration of IgG was detected using an ELISA kit (E10-118, Bethyl Laboratories Inc.).

Calculation of IgG Absorption Efficiency

The apparent efficiency of IgG absorption (AEA) was calculated as described by Quigley and Drewry (1998) using the following formula: $\text{AEA (\%)} = [(\text{serum IgG concentration} \times \text{plasma volume}) / \text{IgG intake}] \times 100$. Plasma volume was estimated as 9.0% of the calf's birth body weight (Quigley et al., 2002). Intake of IgG was calculated by multiplying the IgG concentration of the

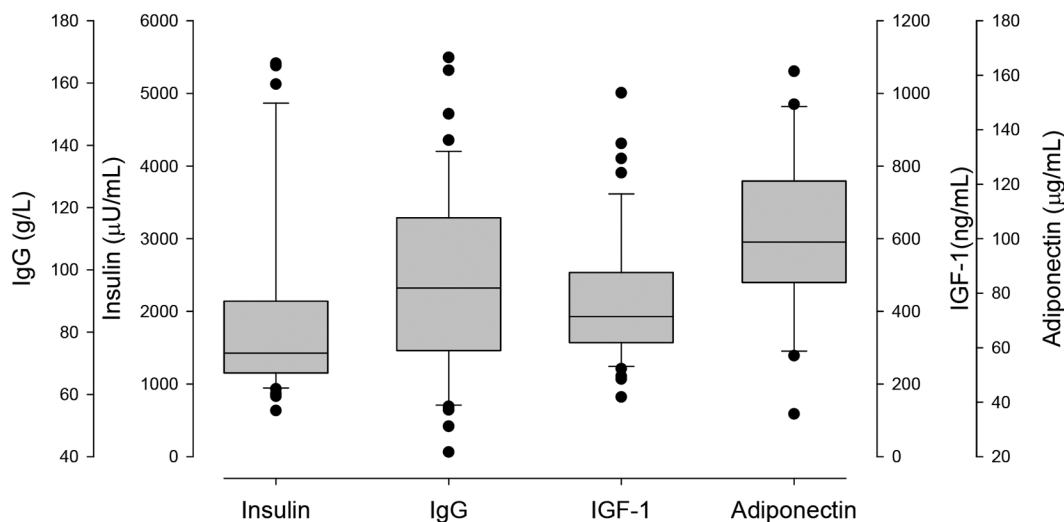


Figure 2. Concentrations of insulin, IgG, IGF-1, and adiponectin in colostrum fed to calves ($n = 46$). The lower and upper edges of the boxes represent the 25th and 75th percentiles, respectively; the midline indicates the median. Whiskers extend to the 10th and 90th percentiles, and dots outside these whiskers denote outliers.

colostrum by the volume ingested during the first feeding (Quigley and Drewry, 1998).

Statistical Analysis

Data presented in the manuscript are mean values $\pm$ SEM. A power analysis (PROC POWER, SAS v9.4, SAS Institute Inc.) based on published data showing metabolic changes in neonatal calves during the first 24 h of life (Hammon and Blum, 1998) was conducted. Primary outcome variables were concentrations of IgG (effect size 41.1 g/L, SD = 14.2 g/L), NEFA (effect size 0.38 mmol/L, SD = 0.25 mmol/L), and glucose (effect size 1.9 mmol/L, SD = 1.2 mmol/L). At $n = 21$ animals, statistical power was estimated >0.99 for the primary outcome variables. Temporal changes in blood parameters over 4, 12, and 24 h p.p. were analyzed using mixed models (PROC MIXED) with age as repeated measure and individual calf as random effect to evaluate overall temporal patterns. To assess the influence of colostrum composition on neonatal blood parameters, additional mixed models were implemented, including colostrum variables (fat, protein, lactose, insulin, adiponectin, colostral IgG) as fixed effects, as well as their interactions with age. Different covariance structures (variance components, compound symmetry, and unstructured) were compared, and the structure providing stable estimates (variance components) was retained, with compound symmetry applied as a sensitivity check for key parameters (IgG, TP, glucose). Pearson correlation coefficients between colostrum variables and changes in blood parameters relative to baseline at 4 h p.p. (Δ 4–12 h and Δ 4–24 h) were calculated using PROC CORR. Statistical significance was set at P

< 0.05 , with $0.05 \leq P < 0.10$ considered a tendency. The reduced sample size at 24 h ($n = 21$) was accounted for in the models and considered when interpreting results. To verify that the subset of calves sampled at 24 h p.p. did not differ systematically from the full data set, baseline characteristics (sex, birth weight, and 4-h plasma variables) were compared between calves with and without 24-h samples. Categorical variables (sex) were analyzed using Fisher's exact test, and continuous variables (birth weight and blood parameters) were compared by independent 2-sample t -tests.

RESULTS

Due to attrition, only 21 calves could be sampled at 24 h p.p., and therefore, to verify that the subset of calves sampled up to 24 h p.p. was representative of the entire study population, comparisons were made between calves with and without 24-h data regarding sex, birth weight, and baseline (4-h) plasma values. Sex distribution did not differ between groups (Fisher's exact test, $P = 1.00$). Likewise, no significant differences were detected for birth weight or any baseline (4-h) plasma variables ($P > 0.70$, 2-sample t -tests). Thus, the 24-h subset was considered representative of the full cohort.

Colostrum Composition

Contents of gross components (i.e., fat, protein, and lactose) and non-nutrient factors (IgG, insulin, IGF-1, and adiponectin) are shown in Figures 1 and 2. All parameters analyzed in colostrum showed considerable variation. It should be noted that adiponectin was mea-

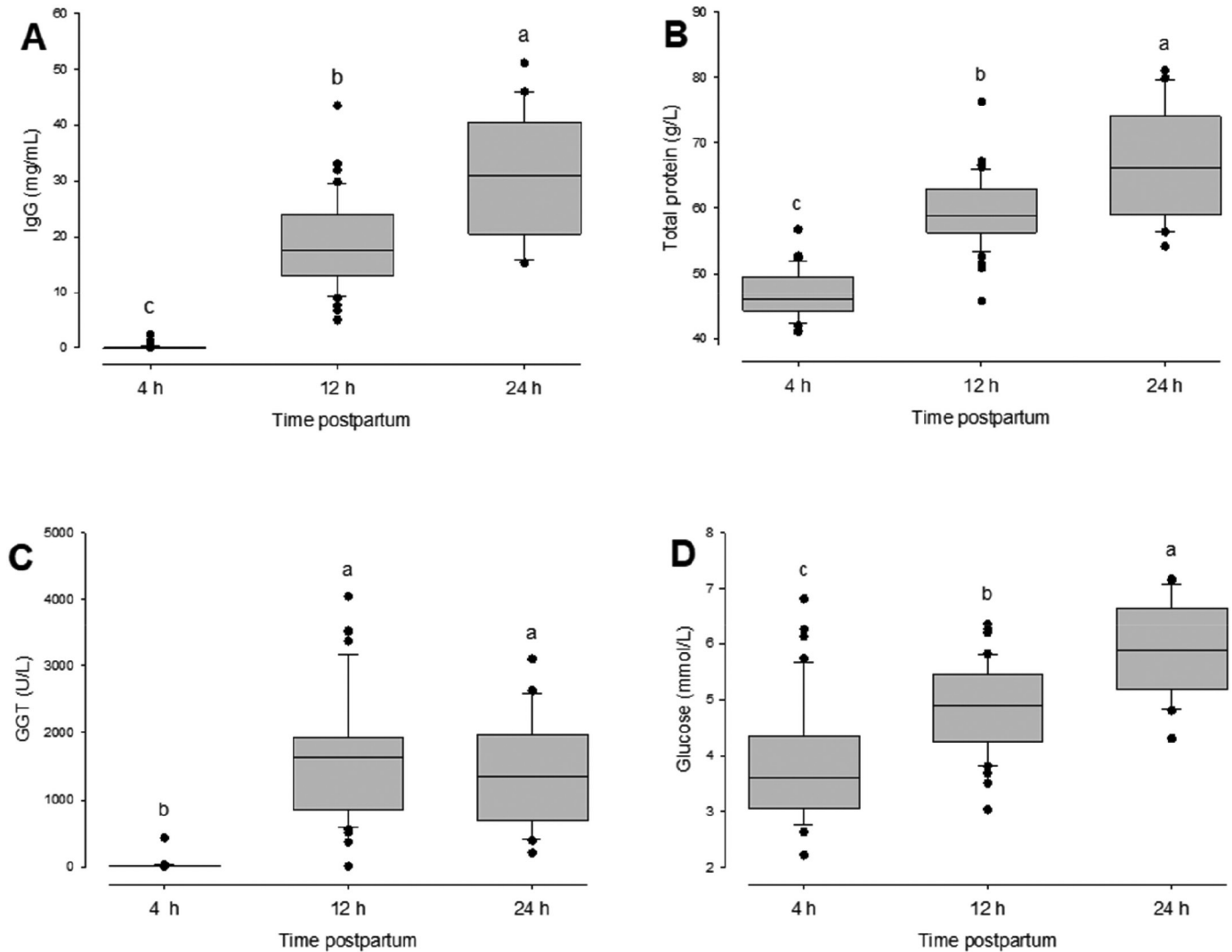


Figure 3. Concentrations of IgG (A) and total protein (B), activity of GGT (C), and glucose concentration (D) in blood of calves at 4 h p.p. (before first colostrum feeding, $n = 46$), 12 h p.p. (8 h after first colostrum feeding and before second feeding, $n = 46$), and 24 h p.p. (12 h after the second colostrum feeding, $n = 21$). Significant differences ($P < 0.05$) between the individual time points are indicated by different letters (a–c). The lower and upper edges of the boxes represent the 25th and 75th percentiles, respectively; the midline indicates the median. Whiskers extend to the 10th and 90th percentiles, and dots outside these whiskers denote outliers.

sured only at 4 and 12 h p.p., due to the limited availability of colostrum samples, and that only 21 calves were available for analysis at 24 h p.p.

Changes in Blood Parameters

The serum concentration of IgG in calves increased after the first colostrum intake ($P = 0.0001$) until 12 h p.p. and further increased after the second colostrum intake ($P = 0.0001$; Figure 3A). At 24 h p.p., serum IgG concentrations were >10 g/L in all animals. The AEA decreased markedly from 12 h to 24 h p.p., with mean $\pm$ SE values of $40.6\% \pm 2.5\%$ and $21.2\% \pm 3.1\%$, respectively ($P < 0.0001$). Similar to IgG concentration,

plasma concentration of TP continuously increased after the first and second colostrum intake ($P = 0.0001$; Figure 3B). As observed with IgG concentration, GGT activity notably increased following the initial colostrum intake ($P = 0.0001$) but did not differ between 12 and 24 h p.p. ($P = 0.3928$; Figure 3C). Glucose concentration in plasma varied widely at 4 h p.p. (Figure 3D). After the first and second colostrum intake, the glucose concentration steadily increased up to 5.83 ± 0.18 mmol/L at 24 h p.p. ($P = 0.0001$).

Immediately after birth NEFA concentration was greatest and decreased until 12 h p.p. ($P = 0.0001$; Figure 4A). Plasma NEFA concentrations were similar at 12 and 24 h p.p. ($P = 0.9986$; Figure 4A). Following the initial

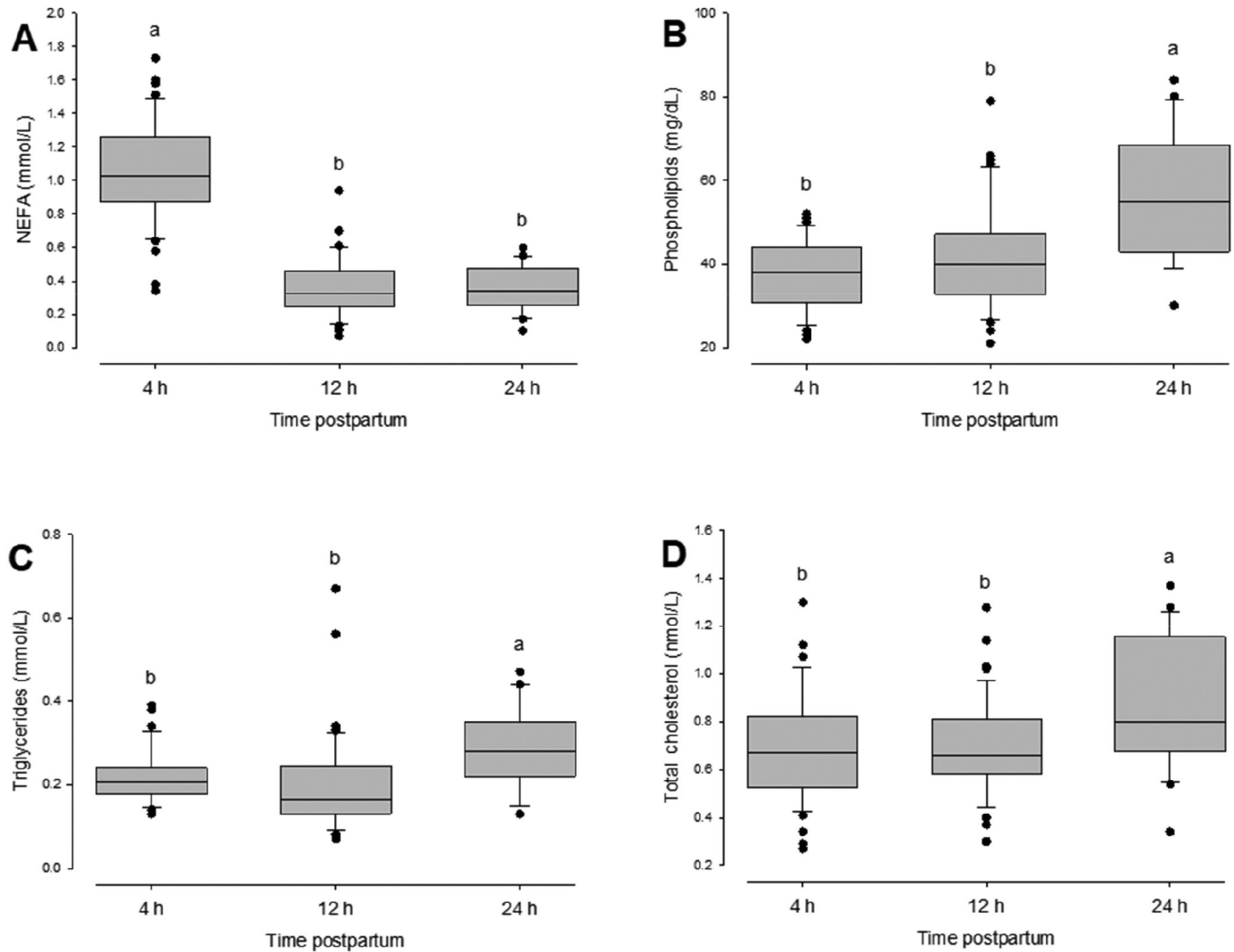


Figure 4. Concentrations of NEFA (A), phospholipids (B), triglycerides (C), and total cholesterol (D) in blood of calves at 4 h p.p. (before first colostrum feeding, $n = 46$), 12 h p.p. (8 h after first colostrum feeding and before second feeding, $n = 46$), and 24 h p.p. (12 h after the second colostrum feeding, $n = 21$). Significant differences ($P < 0.05$) between the individual time points are indicated by different letters (a,b). The lower and upper edges of the boxes represent the 25th and 75th percentiles, respectively; the midline indicates the median. Whiskers extend to the 10th and 90th percentiles, and dots outside these whiskers denote outliers.

colostrum intake, the PL plasma concentration did not change markedly ($P = 0.1665$), but increased thereafter until 24 h of age ($P = 0.0001$; Figure 4B). Concentrations of both TG and TC in plasma did not differ between 4 and 12 h p.p. but were increased at 24 h p.p. (TG: $P = 0.0031$; TC: $P = 0.0165$; Figures 4C and 4D). After the first colostrum intake, an increase in ASAT activity was observed ($P = 0.0001$), which continued to rise after the second colostrum intake ($P = 0.0001$; data not shown).

Serum insulin concentration increased after the first colostrum intake ($P = 0.0001$; Figure 5A) with considerable variation. Following the second colostrum intake, insulin levels remained similar between 12 and 24 h p.p. ($P = 0.1862$; Figure 5A), whereas IGF-1 concentration

remained at a constant level following the initial intake of colostrum ($P = 0.9589$) and exhibited an increase at 24 h p.p. ($P = 0.0436$; Figure 5B). Furthermore, an increase in serum glucagon was observed after the first colostrum intake ($P = 0.0001$), which remained constant after the second feeding ($P = 0.7995$; Figure 5C). The cortisol concentration in serum was elevated until 12 h p.p. and declined thereafter ($P = 0.0042$; Figure 5D).

Associations Between Colostrum Components and Changes in Blood Parameters Up to 12 h p.p.

Table 1 shows Pearson correlation coefficients between individual colostrum parameters and changes in

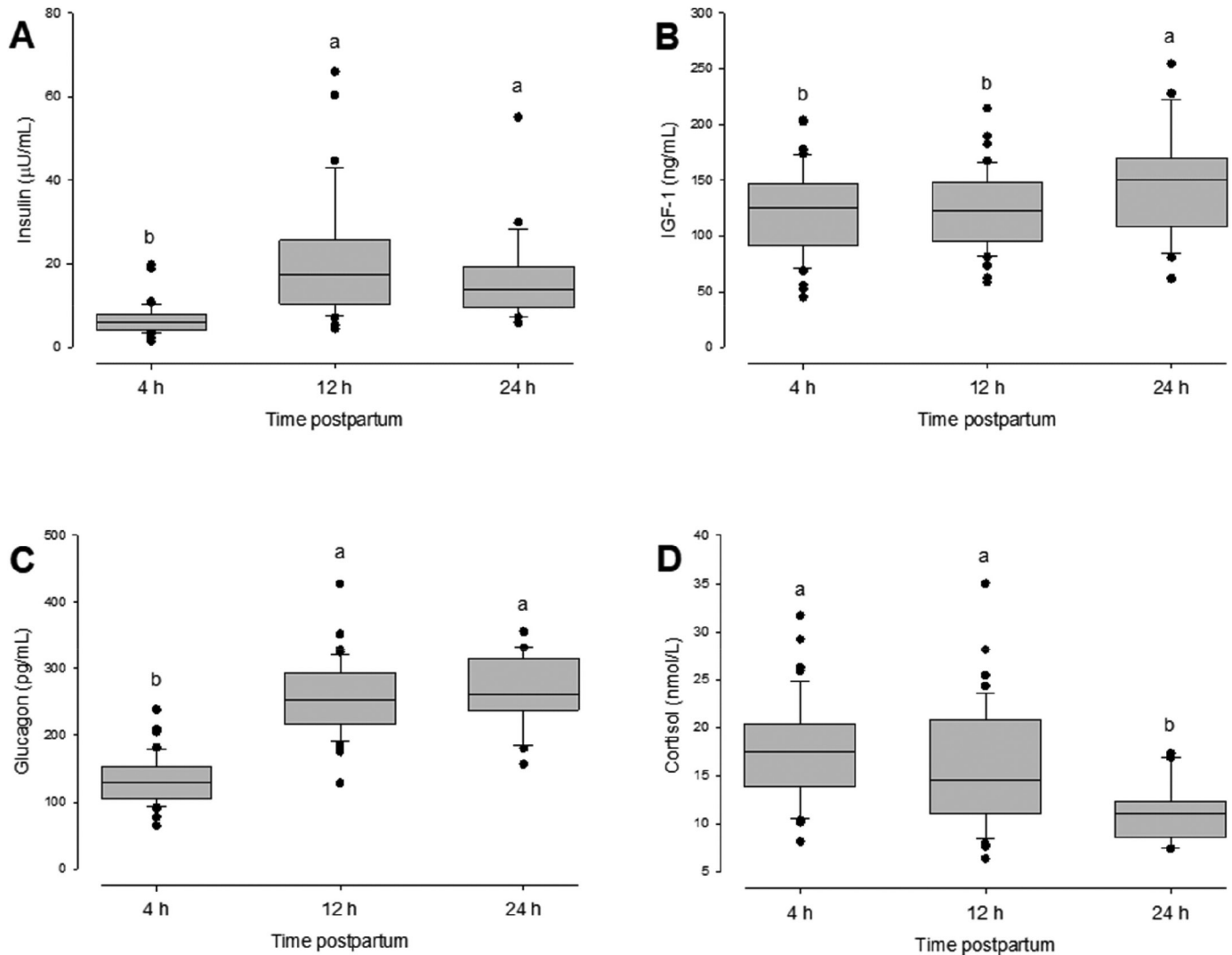


Figure 5. Concentrations of insulin (A), IGF-1 (B), glucagon (C), and cortisol (D) in blood of calves at 4 h p.p. (before first colostrum feeding, $n = 46$), 12 h p.p. (8 h after first colostrum feeding and before second feeding, $n = 46$), and 24 h p.p. (12 h after the second colostrum feeding, $n = 21$). Significant differences ($P < 0.05$) between the individual time points are indicated by different letters (a,b). The lower and upper edges of the boxes represent the 25th and 75th percentiles, respectively; the midline indicates the median. Whiskers extend to the 10th and 90th percentiles, and dots outside these whiskers denote outliers.

blood variables from 4 to 12 h p.p. After the first colostrum intake until 12 h p.p., a strong positive correlation was detected between colostrum fat and PL ($r = 0.95$, $P < 0.0001$), TG ($r = 0.51$, $P = 0.0003$), and TC ($r = 0.51$, $P = 0.0033$), as well as a negative correlation between colostrum fat content and GGT activity in calves ($r = -0.4$, $P = 0.0530$). Furthermore, we found a positive correlation between colostrum fat and change of NEFA concentration ($r = 0.28$, $P = 0.0617$). Additionally, a positive correlation was found between colostrum protein and plasma TP concentration ($r = 0.45$, $P = 0.0017$), whereas protein content in colostrum was negatively associated with changes of PL ($r = -0.29$, $P = 0.0514$) and TC ($r = -0.27$, $P = 0.0644$) from 4 to 12 h p.p. A negative correlation was

observed between colostrum lactose and serum IgG ($r = -0.26$, $P = 0.0766$) as well as TP ($r = -0.45$, $P = 0.0019$). Colostrum IgG and serum IgG ($r = 0.26$, $P = 0.0817$), as well as glucose ($r = 0.27$, $P = 0.0676$) and NEFA ($r = 0.28$, $P = 0.0676$), showed moderate positive correlations (Table 1). Furthermore, a strong positive correlation was identified between colostrum insulin and IgG ($r = 0.52$, $P = 0.0003$), TP ($r = 0.40$, $P = 0.0069$), and glucose ($r = 0.42$, $P = 0.0042$). Conversely, negative correlations were observed with NEFA ($r = -0.27$, $P = 0.0744$), ASAT activity ($r = -0.37$, $P = 0.0841$), and glucagon concentration ($r = -0.27$, $P = 0.0772$). Serum parameters were only poorly correlated with colostrum IGF-1 (Table 1). In contrast, colostrum adiponectin concentration was positively

Table 1. Pearson correlation coefficients between colostrum parameters and changes of variables analyzed in calves from 4 to 12 h of age (n = 46)

Blood parameter	Colostrum parameter						
	Nutrient			Non-nutrient			
	Fat	Protein	Lactose	IgG	Insulin	IGF-1	Adiponectin
IgG	-0.08 (<i>P</i> = 0.61)	0.24 (<i>P</i> = 0.11)	-0.26* (<i>P</i> = 0.08)	0.26* (<i>P</i> = 0.08)	0.52* (<i>P</i> < 0.001)	-0.09 (<i>P</i> = 0.56)	0.37* (<i>P</i> = 0.08)
Total protein	0.05 (<i>P</i> = 0.74)	0.45* (<i>P</i> < 0.01)	-0.45* (<i>P</i> < 0.01)	0.18 (<i>P</i> = 0.24)	0.40* (<i>P</i> < 0.01)	-0.02 (<i>P</i> = 0.90)	0.51* (<i>P</i> = 0.01)
Glucose	-0.01 (<i>P</i> = 0.93)	0.18 (<i>P</i> = 0.23)	-0.17 (<i>P</i> = 0.26)	0.27* (<i>P</i> = 0.07)	0.42* (<i>P</i> < 0.01)	0.06 (<i>P</i> = 0.70)	0.20 (<i>P</i> = 0.36)
NEFA	0.28* (<i>P</i> = 0.06)	0.09 (<i>P</i> = 0.57)	-0.06 (<i>P</i> = 0.68)	0.28* (<i>P</i> = 0.07)	-0.27* (<i>P</i> = 0.07)	0.21 (<i>P</i> = 0.18)	0.17 (<i>P</i> = 0.43)
Phospholipids	0.65* (<i>P</i> < 0.01)	-0.29* (<i>P</i> = 0.05)	0.14 (<i>P</i> = 0.34)	-0.07 (<i>P</i> = 0.66)	0.03 (<i>P</i> = 0.83)	-0.23 (<i>P</i> = 0.12)	0.11 (<i>P</i> = 0.63)
Triglycerides	0.51* (<i>P</i> < 0.01)	-0.19 (<i>P</i> = 0.20)	0.12 (<i>P</i> = 0.43)	-0.07 (<i>P</i> = 0.64)	-0.24 (<i>P</i> = 0.12)	-0.22 (<i>P</i> = 0.15)	0.46* (<i>P</i> = 0.03)
Total cholesterol	0.42* (<i>P</i> < 0.01)	-0.27* (<i>P</i> = 0.06)	0.19 (<i>P</i> = 0.20)	-0.18 (<i>P</i> = 0.24)	-0.07 (<i>P</i> = 0.65)	0.08 (<i>P</i> = 0.59)	-0.35 (<i>P</i> = 0.10)
ASAT	0.09 (<i>P</i> = 0.67)	-0.08 (<i>P</i> = 0.71)	-0.15 (<i>P</i> = 0.48)	0.05 (<i>P</i> = 0.82)	-0.37* (<i>P</i> = 0.08)	0.16 (<i>P</i> = 0.46)	-0.05 (<i>P</i> = 0.81)
GGT	-0.41* (<i>P</i> = 0.05)	0.33 (<i>P</i> = 0.12)	-0.35 (<i>P</i> = 0.10)	0.13 (<i>P</i> = 0.54)	-0.20 (<i>P</i> = 0.38)	0.33 (<i>P</i> = 0.13)	0.17 (<i>P</i> = 0.45)
Insulin	0.15 (<i>P</i> = 0.32)	-0.21 (<i>P</i> = 0.17)	-0.14 (<i>P</i> = 0.37)	-0.14 (<i>P</i> = 0.35)	-0.08 (<i>P</i> = 0.61)	-0.14 (<i>P</i> = 0.39)	0.05 (<i>P</i> = 0.82)
Glucagon	0.13 (<i>P</i> = 0.40)	-0.04 (<i>P</i> = 0.77)	-0.06 (<i>P</i> = 0.68)	-0.10 (<i>P</i> = 0.53)	-0.27* (<i>P</i> = 0.08)	-0.12 (<i>P</i> = 0.42)	0.43* (<i>P</i> = 0.05)
IGF-1	-0.01 (<i>P</i> = 0.94)	0.12 (<i>P</i> = 0.41)	-0.12 (<i>P</i> = 0.43)	-0.05 (<i>P</i> = 0.73)	0.001 (<i>P</i> = 0.95)	0.12 (<i>P</i> = 0.46)	-0.17 (<i>P</i> = 0.43)
Cortisol	-0.06 (<i>P</i> = 0.70)	-0.05 (<i>P</i> = 0.77)	0.16 (<i>P</i> = 0.30)	0.06 (<i>P</i> = 0.70)	-0.26 (<i>P</i> = 0.10)	0.03 (<i>P</i> = 0.85)	0.09 (<i>P</i> = 0.70)

*Significant correlations (*P* < 0.05) and correlations close to significance ($0.05 \leq P < 0.10$).

associated with various parameters, including IgG ($r = 0.37$, $P = 0.0823$), TP ($r = 0.51$, $P = 0.0136$), TG ($r = 0.46$, $P = 0.0271$), and glucagon ($r = 0.43$, $P = 0.0481$).

Associations Between Colostrum Components and Changes in Blood Parameters from 4 to 24 h p.p.

Table 2 shows Pearson correlation coefficients between individual colostrum parameters and changes in blood variables from 4 until 24 h of age. Due to attrition, only 21 calves could be sampled at 24 h p.p., and therefore correlations at this time point are based on this reduced subset. Similar to the associations with physiological changes until 12 h of age, strong positive correlations between colostrum fat and changes of PL ($r = 0.76$, $P < 0.0001$), TG ($r = 0.54$, $P = 0.0122$), and TC ($r = 0.67$, $P = 0.0008$) from 4 to 24 h, and a negative correlation between colostrum fat and circulating IGF-1 ($r = -0.39$, $P = 0.0843$), were observed. For colostrum adiponectin, measurements were only available from cows whose calves were not sampled at 24 h p.p.; consequently, correlations involving adiponectin are restricted to the 4-h and 12-h blood sample data and do not include 24-h p.p. observations. A strong positive correlation was observed between colostrum protein and changes of plasma IgG ($r =$

0.56 , $P = 0.0090$) and TP ($r = 0.49$, $P = 0.0231$), respectively, from 4 to 24 h of age. Colostral lactose, in contrast, correlated negatively with changes of IgG ($r = -0.42$, $P = 0.0603$) and TP ($r = -0.50$, $P = 0.0208$) concentrations. A positive correlation was identified between colostrum IgG and serum IgG ($r = 0.62$, $P = 0.0027$). Additionally, positive correlations between colostrum insulin and alterations of IgG ($r = 0.40$, $P = 0.0731$) and TP ($r = 0.44$, $P = 0.0452$) as well as glucose ($r = 0.48$, $P = 0.0260$) up to 24 h p.p. were identified (Table 2). Concentration of IGF-1 in colostrum and serum cortisol concentration were negatively associated ($r = -0.40$, $P = 0.0720$).

Mixed-Model Analyses Including Colostrum Parameters as Fixed Effects

To further explore the influence of individual colostrum components on blood parameters, mixed models were performed, including colostrum fat, protein, lactose, IgG, insulin, and adiponectin as fixed effects, with calf as a random effect and repeated measures across 4, 12, and 24 h p.p. (Table 3). Due to logistical constraints, only 21 calves could be sampled at 24 h p.p., and consequently only these calves were included in the repeated-measures analyses at this time point. Furthermore, colostrum adipon-

Table 2. Pearson correlation coefficients between colostrum parameters and changes of variables analyzed in calves from 4 to 24 h of age (n = 21)¹

Blood parameter	Colostrum parameter					
	Nutrient			Non-nutrient		
	Fat	Protein	Lactose	IgG	Insulin	IGF-1
IgG	-0.10 (<i>P</i> = 0.67)	0.56* (<i>P</i> < 0.01)	-0.42* (<i>P</i> = 0.06)	0.62* (<i>P</i> < 0.01)	0.40* (<i>P</i> = 0.07)	0.11 (<i>P</i> = 0.63)
Total protein	0.04 (<i>P</i> = 0.86)	0.49* (<i>P</i> = 0.02)	-0.50* (<i>P</i> = 0.02)	0.54* (<i>P</i> = 0.01)	0.44* (<i>P</i> = 0.05)	-0.05 (<i>P</i> = 0.83)
Glucose	0.19 (<i>P</i> = 0.42)	0.28 (<i>P</i> = 0.22)	-0.28 (<i>P</i> = 0.23)	0.37 (<i>P</i> = 0.10)	0.48* (<i>P</i> = 0.03)	-0.12 (<i>P</i> = 0.61)
NEFA	0.13 (<i>P</i> = 0.57)	0.18 (<i>P</i> = 0.44)	-0.37 (<i>P</i> = 0.10)	0.26 (<i>P</i> = 0.26)	-0.30 (<i>P</i> = 0.19)	0.20 (<i>P</i> = 0.39)
Phospholipids	0.76* (<i>P</i> < 0.01)	-0.27 (<i>P</i> = 0.23)	-0.09 (<i>P</i> = 0.70)	-0.12 (<i>P</i> = 0.60)	0.01 (<i>P</i> = 0.95)	-0.09 (<i>P</i> = 0.70)
Triglycerides	0.54* (<i>P</i> = 0.01)	-0.22 (<i>P</i> = 0.34)	-0.20 (<i>P</i> = 0.39)	-0.24 (<i>P</i> = 0.30)	-0.09 (<i>P</i> = 0.70)	0.005 (<i>P</i> = 0.98)
Total cholesterol	0.67* (<i>P</i> < 0.01)	-0.12 (<i>P</i> = 0.61)	0.10 (<i>P</i> = 0.68)	-0.30 (<i>P</i> = 0.19)	-0.11 (<i>P</i> = 0.62)	-0.19 (<i>P</i> = 0.41)
ASAT	0.26 (<i>P</i> = 0.26)	0.07 (<i>P</i> = 0.75)	0.12 (<i>P</i> = 0.60)	0.13 (<i>P</i> = 0.56)	0.08 (<i>P</i> = 0.74)	0.26 (<i>P</i> = 0.25)
GGT	-0.14 (<i>P</i> = 0.53)	0.24 (<i>P</i> = 0.29)	0.01 (<i>P</i> = 0.98)	0.25 (<i>P</i> = 0.28)	-0.08 (<i>P</i> = 0.72)	0.21 (<i>P</i> = 0.37)
Insulin	-0.01 (<i>P</i> = 0.98)	-0.03 (<i>P</i> = 0.91)	0.11 (<i>P</i> = 0.64)	0.16 (<i>P</i> = 0.50)	-0.07 (<i>P</i> = 0.75)	0.07 (<i>P</i> = 0.77)
Glucagon	0.01 (<i>P</i> = 0.97)	0.17 (<i>P</i> = 0.47)	-0.33 (<i>P</i> = 0.14)	0.09 (<i>P</i> = 0.71)	0.14 (<i>P</i> = 0.54)	0.15 (<i>P</i> = 0.50)
IGF-1	-0.39* (<i>P</i> = 0.08)	-0.25 (<i>P</i> = 0.28)	0.32 (<i>P</i> = 0.16)	-0.26 (<i>P</i> = 0.26)	-0.25 (<i>P</i> = 0.27)	0.23 (<i>P</i> = 0.32)
Cortisol	0.12 (<i>P</i> = 0.61)	-0.25 (<i>P</i> = 0.27)	0.30 (<i>P</i> = 0.19)	-0.30 (<i>P</i> = 0.19)	-0.21 (<i>P</i> = 0.35)	-0.40* (<i>P</i> = 0.07)

¹Adiponectin was not measured at 24 h p.p. and is therefore not included in this table.*Significant correlations (*P* < 0.05) and correlations close to significance (0.05 ≤ *P* < 0.10).

nectin concentrations could be measured only in a subset of colostrum samples whose calves were not sampled at 24 h p.p., restricting the adiponectin mixed-model analyses to the 4-h and 12-h data. Colostral fat affected plasma PL (*P* < 0.0001; fat × time: *P* < 0.0001), TG (*P* = 0.0078; fat × time: *P* = 0.0008), and TC concentrations (*P* = 0.0004; fat × time: *P* < 0.0001). Colostral protein influenced serum IgG (*P* = 0.0016; protein × time: *P* = 0.0029) and plasma TP (*P* = 0.0135; protein × time: *P* = 0.0006). Colostral lactose was related to serum IgG (*P* = 0.0074; lactose × time: *P* = 0.0206) and plasma TP (*P* = 0.0086; lactose × time: *P* = 0.0002). Colostral IgG showed effects on calf serum IgG (*P* = 0.0004; IgG × time: *P* = 0.0002) and plasma TP (*P* = 0.0105; IgG × time: *P* = 0.0057). In addition, colostral insulin influenced serum IgG (*P* = 0.0004; insulin × time: *P* = 0.0012) and plasma TP (*P* = 0.0434; insulin × time: *P* = 0.0087) and interacted with time for plasma glucose (*P* = 0.0061). Colostral adiponectin showed tendencies to affect serum IgG (adiponectin: *P* = 0.0818; adiponectin × time: *P* = 0.0823) and interacted with time for plasma TP (adiponectin × time: *P* = 0.0136), but analyses were limited to 4-h and 12-h samples due to the absence of

corresponding 24-h data. No effects of colostral components were detected on plasma IGF-1 concentrations. These mixed-model results are largely consistent with the observed correlation analyses, supporting the robustness of the associations between colostral components and blood parameters.

DISCUSSION

Freihofer et al. (2025) showed in a companion paper that the maternal metabolic and health status immediately after birth partly affected neonatal calves' metabolism until 24 h p.p. independent of colostrum feeding. Despite the additional environmental impact, colostrum and its bioactive components are very likely to affect metabolic adaptation in neonatal calves. However, limited knowledge exists on the role of variation in individual colostrum components on metabolic and endocrine adaptation in calves. Therefore, the objective of this study was to evaluate the influence of colostral nutrient and non-nutrient factors on physiological parameters of calves fed colostrum from their own dam within the first 24 h of age.

Table 3. Effects of colostral components on calf blood parameters (PROC MIXED with repeated measures; fixed effects: colostral component, time, and interaction [colostral component $\times$ time])

Colostrum component	Blood parameter	Component effect (<i>P</i> -value)	Time effect (<i>P</i> -value)	Interaction (<i>P</i> -value)
Fat	Phospholipid	<0.0001	0.0003	<0.0001
	Triglyceride	0.0078	0.0004	0.0008
	Total cholesterol	0.0004	0.0223	<0.0001
Protein	Serum IgG	0.0016	0.4284	0.0029
	Total protein	0.0135	0.3912	0.0006
Lactose	Serum IgG	0.0074	<0.0001	0.0206
	Total protein	0.0086	<0.0001	0.0002
	Glucose	0.8980	0.0485	0.4728
IgG	Serum IgG	0.0004	0.0060	0.0002
	Total protein	0.0105	0.0032	0.0057
Insulin	Serum IgG	0.0004	<0.0001	0.0012
	Total protein	0.0434	<0.0001	0.0087
	Glucose	0.7728	0.0004	0.0061
Adiponectin	Serum Insulin	0.9847	0.0040	0.7532
	Serum IgG	0.0818	0.1245	0.0823
	Total protein	0.4907	0.0603	0.0136
	Triglyceride	0.3528	0.0086	0.0271
	Glucagon	0.9101	0.4131	0.0424

Our results confirm earlier findings on the variation of individual colostral components despite identical housing and feeding (Grosvenor et al., 1993; Baumrucker et al., 2010; Kessler et al., 2020). Fat and protein contents in colostrum samples of the present study were in line with observations from Quigley and Drewry (1998) and Kessler et al. (2020). Fat represents the most important energy source in colostrum for the calf (Hammon and Blum, 1998) and is critical for GNG because the low lactose content in colostrum is not sufficient to provide enough energy to the calf (Hammon and Blum, 1998; Silva et al., 2024). Our data also show a strong correlation between metabolizable energy and later lipid plasma concentrations (data not shown). Even though fatty acids are readily absorbed as an energy source (Fischer-Tlustos et al., 2021), TC, TG, and PL concentrations in the present study were increased only at 24 h p.p. One explanation for the delayed appearance of lipids in plasma is provided by the findings of Miyazaki et al. (2017). According to their observations, proteins and lipids in colostrum form a curd that can retain these nutrients and release them more gradually into the intestine, thereby delaying absorption. In their study, a distinct curd structure was still detectable by ultrasonography up to 8 h after first colostrum ingestion. However, other studies using acetaminophen absorption or scintigraphy have reported considerably faster emptying rates of milk or colostrum (Burgstaller et al., 2017; Desjardins-Morrisette et al., 2018). Though abomasal emptying or curd formation were not measured in the present study, we can speculate that these factors may partly explain our findings. Considering the elevated NEFA levels before the first colostrum intake, which decrease concomitantly to the simultaneously rising glucose concentrations, we assume that the calf covers its

energy demand during the first hours of life from GNG based on mobilized body reserves as well as from the steadily increasing intestinal glucose absorption. Initially high, but then decreasing after colostrum intake, NEFA plasma concentrations have also been observed in other studies (Egli and Blum, 1998; Kühne et al., 2000; Piccione et al., 2010). Elevated NEFA concentrations occur due to the increased mobilization of fat tissue (Li et al., 2013). Steinhardt et al. (1995) attributed the high NEFA levels to the consequence of the increased sympathomedullary activity of the newborn calf, resulting in increased lipolysis. As expected, mixed-model analyses confirmed that colostral fat affected plasma concentrations of TC, TG, and PL, with pronounced time-dependent interactions (fat $\times$ time). These findings corroborate the correlation analyses and suggest that colostral fat contributes to early postnatal lipid metabolism and influences the temporal dynamics of circulating lipid concentrations during the first 24 h p.p., although causal mechanisms remain to be established.

After fat, protein in colostrum is the second most important dietary energy source. Total protein concentration in plasma has been shown to be positively associated with immunoglobulin content of colostrum (Butler and Kehrl, 2005; Løkke et al., 2016; Dunn et al., 2017). Our observations confirmed strong correlations between colostral protein intake and plasma TP after the colostrum intake. Interestingly, we found no significant correlation between colostral protein and plasma IgG concentration after the first colostrum intake. Similarly, we observed only a poor correlation between colostral IgG and plasma TP concentration at 12 h of age. Nevertheless, we recorded strong positive associations between colostral protein and IgG with circulating TP and IgG concentra-

tions at 24 h p.p. These associations were also supported by the mixed-model analyses, which showed significant main effects of colostrum protein and colostrum IgG on plasma TP and IgG, including interactions with time. We assume that the appearance of plasma IgG relative to changes in TP concentration is due to the absorption of protein components by pinocytosis (Blum, 2006) that are initially independent of the colostrum IgG content. After first colostrum intake, the absorption capacity of the small intestine could be reached, and a positive correlation only emerges as IgG absorption continues. The generally strong correlation between these parameters is well documented and corroborated by other authors (Naylor and Kronfeld, 1977; Tyler et al., 1996). Furthermore, our data are in alignment with findings of Girard et al. (1992) and Hammon and Blum (1998), describing the relationships between the amount of IgG in colostrum and the subsequent plasma concentration. Our data demonstrate a notable and almost linear increase in plasma IgG and TP concentration over the initial 24-h period, as earlier shown by Rauprich et al. (2000). Similarly, GGT, which is also absorbed via colostrum (Thompson and Pauli, 1981; Gregory, 2003; Mortazavi et al., 2024), is employed to assess colostrum intake and frequently serves as an indicator of successful immunization and IgG absorption (Hogan et al., 2015). In contrast to IgG, which continued to increase in a linear fashion following the first and second intake of colostrum, GGT activity in calves was below detection limit (or only very low) before first colostrum intake, but increased markedly to constant levels until 24 h p.p.

After birth, calves experience a hypoglycemic state (Girard et al., 1992; Hammon and Blum, 1998; Hammon et al., 2013), which is evidenced by the initially high plasma NEFA and low glucose concentrations before the first colostrum intake (Ronge and Blum, 1988). As previously mentioned, GNG plays a pivotal role in maintaining glucose and therefore energy homeostasis in the neonatal calf (Blum and Hammon, 2000). In contrast, GNG activity remains relatively low in the postnatal calf (Hammon et al., 2012). However, energy reserves of the newborn calf are assumed to last for a period of only 3 h postnatally (Okamoto et al., 1986). In addition to eGP, lactose in colostrum represents another important energy source for the calf (Lopez and Heinrichs, 2022). Fowden et al. (2009) indicated that fetal glucose concentration largely depends on the placental support provided by the dam. Freihofer et al. (2025) showed only a moderate correlation of maternal glucose with glucose concentration in calves after parturition. Nevertheless, a notable elevation in plasma glucose levels was discerned following the initial as well as the subsequent colostrum intake. This finding is consistent with observations made by numerous other studies (e.g., Stanko et al., 1992; Hammon and

Blum, 1998). As investigated by Scheuer et al. (2006) and Hammon et al. (2013), eGP, including GNG, contributes significantly to glucose homeostasis during the first few days. However, the delayed appearance of absorbed lipids in circulation and the rapidly decreasing NEFA concentrations after the first colostrum intake suggest that a notable proportion of plasma glucose originates from enzymatically cleaved lactose. Interestingly, however, we observed no significant association between colostrum lactose and plasma glucose concentration, which may be due to the influence of eGP, as lactose cleavage is not sufficient to compensate for the glucose deficiency (Hammon et al., 2013). However, negative correlations between colostrum lactose and plasma IgG and TP concentrations were observed as tendencies, whereas mixed-model analyses revealed statistically significant associations. Earlier research in piglets showed that lactose and its monosaccharides reduce Ig absorption (Lecce, 1966; Werhahn et al., 1981).

It has been established that colostrum contains a considerable number of bioactive substances. In our study, we determined the presence of bioactive substances in colostrum, including insulin, adiponectin, and IGF-1, and evaluated their influence. Insulin concentration is present at markedly elevated concentrations in colostrum compared with mature milk (Malven et al., 1987; Grosvenor et al., 1993). The extent to which colostrum insulin may be absorbed or affect plasma concentrations in calves is a subject of considerable debate. Grütter and Blum (1991) demonstrated that insulin is poorly absorbed, which confirms our present results showing a correlation between colostrum and plasma insulin concentrations close to zero. Although insulin and glucagon are known to be antagonists, both insulin and glucagon concentrations increased following the initial intake of colostrum in the present study. This finding suggests a certain independence between insulin and glucagon in the neonatal calf, at least in the first hours postpartum. An explanation for the apparently independent increase of glucagon in regard to GNG could be a greater sympathetic activity (Steinhardt et al., 1995). Mixed-model analyses revealed positive effects of colostrum insulin on serum IgG and TP concentrations, consistent with the observed correlations. To the best of our knowledge, no similar observations have been reported in literature so far. Given the evidence that insulin plays a role in postnatal villi height and mucosal surface growth (Fischer et al., 2019; Hare et al., 2023), it is plausible to hypothesize that colostrum-derived insulin might contribute to enhanced IgG and protein absorption. However, no histological or morphological data were collected in this study, and this remains a speculative, hypothesis-generating interpretation. Furthermore, a positive correlation was observed between colostrum insulin and plasma glucose concentration, although no

significant relationship was observed in the mixed-model analysis. Hare et al. (2023) showed that insulin supplementation in neonatal calves increased the enzymatic activity of lactase in the proximal jejunum and enabled an enhanced glucose absorption.

Similar to insulin, adiponectin is present in high concentrations in colostrum and probably absorbed in larger quantities (Kesser et al., 2015). Although we did not determine adiponectin concentration in calves, a positive correlation was demonstrated between colostrum and blood adiponectin concentration by Kesser et al. (2017). It should be noted that colostrum adiponectin concentrations were only measured in 21 calves, and no measurements were available at 24 h p.p. due to the unavailability of blood samples. Consequently, the mixed-model and correlation analyses for colostrum adiponectin are limited to the 4-h and 12-h data, which should be considered when interpreting the observed associations. Adiponectin is widely recognized for its enhanced sensitivity to insulin (Berg et al., 2001; Fruebis et al., 2001; Yamauchi et al., 2001). Similarly, adiponectin may also exert a beneficial influence on intestinal development (Kesser et al., 2015). In the mixed-model analyses, colostrum adiponectin showed tendencies to affect serum IgG, in agreement with the correlations observed at 4 and 12 h p.p. These findings are intended to provide preliminary insights and should be interpreted as a basis for future investigation rather than as definitive evidence. Given the well-documented sensitivity-enhancing effect of adiponectin on insulin, it seems plausible to suggest that our observations are related to the associations between colostrum insulin and plasma concentrations of IgG. It could be hypothesized that adiponectin may have a local effect on insulin sensitivity within the intestine, which could lead to enhanced intestinal development or an increase in the mucosal surface area, contributing to improved IgG absorption. However, no histological or morphological data were collected to confirm this interpretation, and these considerations remain speculative. Additionally, further positive correlations were identified between colostrum adiponectin and plasma concentrations of glucagon and TG. However, these associations were not confirmed as significant in the mixed-model analysis, indicating that they should be interpreted with caution. In contrast to our findings, adiponectin has been shown to reduce GNG (Miller et al., 2011; Stern et al., 2016). Moreover, adiponectin enhances the clearance of fatty acids and TG by augmenting fatty acid oxidation in the liver (Combs et al., 2004). Our contradictory findings might indicate that these physiological relationships may not yet be established in the neonatal calf. Thus, impaired GNG could be preventable during the first hours of life despite increasing adiponectin concentrations.

In contrast to adiponectin, IGF-1 is poorly absorbed and recognized to contribute to intestinal development (Vacher et al., 1995; Hammon et al., 2000), as confirmed by the low correlation between colostrum and serum IGF-1 concentrations in our study. Apart from a negative correlation with cortisol, as shown earlier (Cianfarani et al., 2002; Ricotti et al., 2020), no notable associations with serum parameters were observed in the present study. The lack of clear relationships between colostrum IGF-1 and systemic variables, despite its well-known trophic role, may be due to the short observation period and the limited absorption of colostrum IGF-1. It has been suggested by previous studies that the trophic effects of IGF-1 on villus morphology and intestinal enzyme activity may require longer exposure than the early postnatal period examined here, as sustained IGF-1 supplementation over several days has been associated with villus growth and enhanced enzymatic activity (Burrin et al., 1996). Furthermore, it is conceivable that, similar to observations in neonatal rodents, intestinal macrophages in calves may locally produce IGF-1 during the early postnatal period (Yan et al., 2022). Such local production could contribute to mucosal and microvascular development and thereby obscure potential associations between colostrum and systemic IGF-1 concentrations. In addition, postabsorptive regulatory mechanisms, including IGF-binding proteins (**IGFBP**), are known to modulate IGF-1 bioavailability and receptor interaction, influencing its local activity independently of systemic concentrations (Clemmons, 2016). Therefore, local IGFBP regulation in the neonatal intestine could further modify IGF-1 action and mask potential relationships with colostrum IGF-1. Overall, these interpretations remain speculative, highlighting the need for future studies combining morphological and molecular analyses to elucidate the local regulation and functional role of IGF-1 in the neonatal calf intestine.

CONCLUSIONS

The substantial shift in nutritional physiology of the neonatal calf is reflected in occasional considerable alterations in the concentrations of measured serum parameters. Colostrum supplies both macronutrients and bioactive factors that appear to shape early metabolic and immunological adaptation. In our dataset, colostrum fat was strongly associated with increases in plasma phospholipids, triglycerides, and total cholesterol. Colostrum IgG was positively associated with calf serum IgG concentrations. By contrast, higher colostrum lactose tended to be associated with lower serum IgG and total protein, and colostrum adiponectin showed a tendency toward a positive relationship with IgG, although these latter observations are less consistent. Taken together, these re-

sults suggest that individual colostrum constituents may modulate early nutrient and immunoglobulin dynamics.

NOTES

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Nonstandard abbreviations used: AEA = apparent efficiency of IgG absorption; ASAT = aspartate aminotransferase; eGP = endogenous glucose production; GGT = gamma-glutamyl transferase; GIT = gastrointestinal tract; GNG = gluconeogenesis; IGF-1 = insulin-like growth factor-1; IGFBP = IGF-binding proteins; NEFA = nonesterified fatty acids; PL = phospholipids; p.p. = postpartum; TC = total cholesterol; TG = triglycerides; TP = total protein.

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